

A proper exercise (ipad)

Given the following corpus:

$\langle S \rangle$ i saw a cat $\langle \backslash S \rangle$
 $\langle S \rangle$ i saw a dog $\langle \backslash S \rangle$
 $\langle S \rangle$ i saw the dog $\langle \backslash S \rangle$
 $\langle S \rangle$ i jumped $\langle \backslash S \rangle$
 $\langle S \rangle$ a cat jumped high $\langle \backslash S \rangle$
 $\langle S \rangle$ the dog barked loud $\langle \backslash S \rangle$
 $\langle S \rangle$ a cat jumped high $\langle \backslash S \rangle$
 $\langle S \rangle$ the cat slept $\langle \backslash S \rangle$

Under a **bigram** model, compute:

1. $p(\text{the} | \langle S \rangle) \rightarrow P(\langle S \rangle \text{the cat jumped})$
- ✓ 2. $p(\text{the cat barked})$
3. $p(\text{the dog barked})$
4. $p(\text{i saw the dog})$

VOCAB (N=13)	
$\langle S \rangle$	dog
$\langle \backslash S \rangle$	the
i	jumped
saw	high
a	barked
cat	loud
slept	

$$\begin{aligned}
 & 1. \quad p(\text{the} | \langle S \rangle) \times p(\text{cat} | \text{the}) \times p(\text{jumped} | \text{cat}) \\
 & = \frac{C(\langle S \rangle, \text{the})}{C(\langle S \rangle)} \times \dots
 \end{aligned}$$

$$= \frac{2}{84} \times \frac{1}{3} \times \frac{2}{42} = \boxed{\frac{1}{24}}$$

$$2. \quad p(\text{the} | \langle S \rangle) \times p(\text{cat} | \text{the}) \times p(\text{barked} | \text{cat})$$

$$= \frac{2}{8} \times \frac{1}{3} \times 0 = 0$$

$$4. \quad p(\langle S \rangle | \text{saw the dog})$$

$$\begin{aligned}
 & p(i | \langle S \rangle) \times p(\text{saw} | i) \times p(\text{the} | \text{saw}) \\
 & \times p(\text{dog} | \text{the})
 \end{aligned}$$

$$= \frac{1}{8} \times \frac{3}{4} \times \frac{1}{3} \times \frac{2}{3}$$

$$= \frac{1}{12}$$